Project 3 Proposal: Lottery vs. Fixed Bonuses: Does Probability Weighting Buy More Effort from Crowd Workers?

Team Katseye: Kevin Lei (kl2344), Kyungmin Nam (kn467), Brittany Sun (bs835)

Psychological factor: probability weighting

People do not treat probabilities in proportion to their size. In prospect theory (Kahneman and Tversky), decisions run through a probability weighting function that overweights small probabilities and underweights moderate-to-large ones. This is why people buy lottery tickets and insurance in the same week: a 1-in-20 chance of a prize feels larger than 5%, so a gamble can feel more attractive than a sure payment of the same expected value.

Expected effect on worker performance

We expect a lottery bonus to buy more effort than a fixed bonus with the same expected value. Requesters commonly pay a small fixed bonus for accurate work. If workers have a small chance at a larger prize, paying that same budget out as a lottery should make the bonus feel bigger and push workers to work more carefully.

Concretely, we predict that workers in the lottery condition will spend more time per item and make fewer counting errors than workers in the fixed-bonus condition. If true, a requester could raise work quality without raising cost. We hold this prediction loosely: lottery incentives are also riskier, and some workers may discount a prize they are unlikely to win, which would push the effect toward zero.

Experiment

We will run a two-condition, between-subjects study on Prolific in which the only difference between conditions is how the accuracy bonus is paid out.

Question and expected answer

Question: when the expected bonus is held equal, does a lottery bonus produce more careful work than a fixed bonus on an effort-driven counting task?

Expected answer: Yes. Lottery-condition workers will count more grids exactly right, have lower average counting error, and spend more time per grid.

Task

Each worker sees 12 grids of digits, 10 rows by 10 columns, one grid per page, and answers "How many 7s are in this grid?" The non-target digits are mostly 1s and 4s, which look similar to 7 and force careful scanning. Each grid holds between 8 and 20 sevens. We chose this task, adapted from the counting tasks used in experimental economics (Abeler et al., 2011), because accuracy depends on how carefully people count, not on skill or prior knowledge. That makes it sensitive to incentives.

Source of data

We will generate all grids ourselves with a Python script, so the correct count for every grid is known exactly. Every worker sees the same 12 grids in a randomized order.

Conditions

All workers receive the same base pay and the same bonus threshold: at least 10 of 12 grids counted exactly right. The bonus terms are shown on the instructions page and repeated on every task page.

Condition	Bonus for meeting the threshold	Expected value
Fixed	$0.25 paid to every qualifying worker	$0.25
Lottery	One $4 prize drawn at random among all qualifying workers (about 1 in 15 if half qualify)	about $0.27

The lottery is real: we will draw a winner and pay the prize through Prolific's bonus system. Participants are told the exact rule, not invented odds.

Procedure

1. Workers accept the study on Prolific and are sent to a Qualtrics survey, which randomly and evenly assigns them to a condition.
2. They read instructions and the bonus terms for their condition, then answer one comprehension question about how the bonus works (our manipulation check).
3. They complete the 12 grids. Qualtrics records each answer and the time spent on each page.
4. One very easy grid serves as an attention check.

Measures and analysis

- Number of grids counted exactly right (out of 12)
- Absolute counting error per grid
- Time spent per grid, in seconds

We will compare conditions with a mixed-effects model on the per-grid data (about 360 grids per condition), with a random intercept per participant, and report a simple per-person comparison alongside it.

Sample and budget

We plan 30 participants per condition (60 total), at an estimated 5 minutes per person and $0.67 base pay, which is Prolific's $8/hour minimum.

Item	Cost
Base pay: 60 × $0.67	$40.20
Fixed bonuses: about 15 × $0.25	about $3.75
Lottery prize	$4.00
Prolific platform fee (33.3%)	about $16.00
Total	about $64

We will soft-launch with 10 participants (5 per condition) to check completion time and score spread before releasing the remaining 50 places. If nothing changes, those 10 stay in the dataset.

Limitations

With 30 per condition we can reliably detect only fairly large effects (around d = 0.75). A null result will therefore not show that lotteries are ineffective, only that any effect is smaller than we can detect. The bonus threshold is also set without a full pilot. If almost everyone, or almost no one, qualifies, the bonus stops mattering and both conditions will look the same.